

SHANMUGA MANI M

AI/ML Engineer & Creative Developer

Email: shanmugamani.al23@krct.ac.in | Location: India

PROFESSIONAL SUMMARY

Specialized in building intelligent applications, immersive web experiences, and high-performance digital systems. Focused on combining artificial intelligence, modern development frameworks, and cinematic interface design to create products that are technically robust and visually impactful.

TECHNICAL SKILLS

Languages & Tech: Python, JavaScript, TypeScript, HTML/CSS

AI/ML & Vision: TensorFlow, Keras, OpenCV, Scikit-learn, XGBoost, MediaPipe, LLMs (Groq/Gemini)

Web & 3D: React, Next.js, Node.js, FastAPI, Flask, Three.js (WebGL), Tailwind CSS

Databases: PostgreSQL, MongoDB

SELECTED PROJECTS

VORA AI FITNESS APP (2025)

AI / Full Stack / Fitness

AI-based fitness application for tracking workouts and monitoring activities using smart recommendations. Built with React, Node.js, Flask, TensorFlow, and OpenCV.

Gesture Detection Text-to-Speech (2025)

AI / Computer Vision

Detects hand gestures in real time via webcam and converts them into text and speech, bridging communication gaps. Implemented using Python, OpenCV, and deep learning (TensorFlow/Keras).

Spatial Interaction Workspace (2025)

3D / Gesture / WebGL

Web-based 3D workspace enabling real-time gesture-based object manipulation and scene building. Features intuitive hand tracking to control 3D objects using Three.js and MediaPipe.

FDIA Medical Security System (2025)

ML / Cybersecurity

Machine learning-based solution to detect False Data Injection Attacks (FDIA) in medical sensor data using XGBoost with 98.4% accuracy, ensuring patient data integrity.

AI Fitness Chatbot (2025)

Full Stack / AI / LLM

Full-stack AI fitness chatbot delivering personalized workout plans and real-time conversations, powered by FastAPI, React, and Groq/Gemini API.

Online Learning Platform (2025)

Full Stack / EdTech

Scalable digital environment designed to facilitate, manage, and deliver educational content. Built on a Flask/Node.js

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backend with PostgreSQL.

CERTIFICATIONS (Selected from 23+)

- Master in Data Scientist (IBM / Cognitive Class)
- Cyber Security Expert (Security Specialist)
- Machine Learning and Deep Learning (NPTEL)
- Microsoft Azure Fundamentals
- CompTIA Security+ & Ethical Hacker